MIL-R-24037/4A(SHIPS,
9 June 1967
SUPERSEDING
MIL-R-24037/4(SHIPS)
20 May 1964
(see 6.2)

MILITARY SPECIFICATION

RADIACMETER - IM-135( )/PD

1. SCOPE

1.1 This specification covers an ion-chamber quartz-fiber dosimeter (radiacmeter) capable of measuring and indicating doses of gamma or X-radiation over the range of 0 to 5 Roentgens (R).

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-R-24037 - Radiacmeter, Quartz-Fiber, Self-Indicating, General Specification.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The equipment shall be in accordance with MIL-R-24037 and as specified herein.

3.2 Scale.- The scale shall have 25 equal divisions, each division representing 0.2 Roentgens. There shall be 6 major marking lines, at 0-1-2-3-4-5 Roentgens. The major marking lines shall be marked with arabic numerals.

3.3 Range.- The range of the dosimeter shall be 0 to 5 Roentgens.

FSC 6665

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3.4 Leakage.- The electrical resistance between the charging electrode and the barrel shall be such that leakage in the absence of radiation shall be not more than 0.05 R per day for any relative humidity to 100 percent, and at any temperature between minus 65° and plus 131°F. Leakage shall be taken over a period of 5 days. The initial reading shall be taken 8 to 24 hours after charging. The final reading shall be taken between 108 and 120 hours after initial reading.

3.5 Leakage after heavy irradiation.- After a continuous or intermittent exposure totaling 2000 Roentgens (R) given at dose rates up to 1200 R per hour, electrical leakage in a four hour period starting not more than 10 minutes after exposure is completed, shall not exceed 5 percent of full scale. After a 48-hour period, the leakage requirement specified in 3.4 shall apply.

3.6 Accuracy.- The dosimeter shall measure true gamma dose from a cobalt 60, radium, or cesium 137 source within plus or minus 10 percent or plus or minus 0.13 Roentgen, whichever is larger. This accuracy requirement shall apply for any source which has been:

(a) Calibrated directly by the National Bureau of Standards.
(b) Calibrated by the contractor using Victoreen R meters, or equivalent approved by the command or agency concerned, which have in turn been calibrated by the National Bureau of Standards.
(c) Calibrated by a Government activity using Victoreen R meters, or equivalent approved by the command or agency concerned, which have been calibrated by the National Bureau of Standards.

Specifically included in this spread are the usual imprecisions involved in correlating source strengths with the standard National Bureau of Standards sources.

3.7 Energy dependence.- Except at the energies represented by cesium 137, and cobalt 60, where the requirements of 3.6 shall apply, the dosimeter shall indicate true gamma dose within plus or minus 15 percent of 0.15 Roentgens, whichever is larger, for X- or gamma radiation of any energy from 80 thousand electron volts (KEV) to the energy of cobalt 60 gammas. Over the range 80 KEV to 50 KEV, the dosimeter response shall decrease smoothly; at 50 KEV and at all X- and gamma radiation energies below 50 KEV the response shall be not greater than 50 percent of true dose. As used here, 80 KEV and 50 KEV refer to effective energies of X-ray beams from a constant voltage X-ray machine, with appropriate tube voltage and filtration in accordance with normal National Bureau of Standards procedures.

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3.8 Ion chamber saturation and rate dependence.- The dosimeters, when subjected to dose rates up to 5000 R per hour shall not vary from their normal calibration accuracy by more than plus or minus 0.25 R.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be as specified in MIL-R-24037, and as specified herein.

4.2 Leakage.- The dosimeter shall be fully charged and tested for a period of 5 days at an ambient temperature of at least 100°F. The deflection of the electroscope fiber shall not exceed 0.25 Roentgen during this test. The dosimeter may be stored under charge for some time prior to this test if desired. The fiber shall be charged positively in this and all other tests.

4.3 Calibration at midpoint only.- A source, calibrated in accordance with 3.6 shall be used. Temperature may be normal room temperature. A dose of 2.5 Roentgens shall be delivered to the dosimeter. The reading shall not be greater than 2.75 R nor less than 2.25 R.

4.4 Calibration over the entire range.- The instrument shall be tested as specified in 4.3 at five points: 1 R, 2 R, 3 R, 4 R and 5 R. Readings shall be as specified in 3.6.

5 PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-R-24037.

6. NOTES

6.1 The notes specified in MIL-R-24037 are applicable to this specification.

6.2 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 6665-0192)

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